Class 6 SP250: Counting Part 1

Independent choices:

Question: Tanya can have either bagel or eggs for breakfast and either sandwich, pasta or salad for lunch. How many ways are there for Tanya to have her meals today?

Answer: 2 choices for breakfast and 3 choices for lunch.

FOR EACH of her 2 breakfast choices, she has 3 choices for lunch.

Hence Tanya has 2 * 3 = 6 choices total.

Question: Tanya can have either bagel or eggs for breakfast, either sandwich, pasta or salad for lunch and either beef, lamb or chicken for dinner. How many ways are there for Tanya to have her meals today?

Answer: 2 choices for breakfast, 3 choices for lunch and 3 choices for dinner.

FOR EACH of her 2 choices for breakfast, Tanya has (3 * 3 = 9) choices for lunch and dinner.

Hence the total number of choices is 2 * 3 * 3 = 18.

Permutations:

There are n students in a class. How many ways are there to arrange the students in a line?

There are $n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot 2 \cdot 1 = n!$ ways to do so:

n choices for the first student in a line.

(n-1) choices for the second student.

…. so on.

There are n! ways to permute n distinguishable objects. This is pronounced n-factorial.

By convention, 0! = 1.

Chess rooks and permutations:

Question: How many ways are there to arrange 8 rooks on a chessboard so that no two attack each other?

Answer: 8!. Because there are 8 rows, there must be exactly one rook in each row. There are 8 ways to put a rook in the first row, 7 ways to put a rook in the second row because it has to be in a different column from the first rook, 6 ways to put a rook in a third row, etc.

Total = 8 * 7 * 6 * … * 2 * 1 = 8! ways.

Question: How many ways are there to put a red rook and a green rook on a 4x4 chessboard so that they do not attack each other?

Answer: there are 16 ways to put a green rook and, after it's placed, **for each of these 16 choices**, there are 9 ways to put a red rook down. There are a total of 16 * 9 = 144 ways to put rooks in such a way.

Question: 10 runners are competing in a race. How many possible ways are there for the medals to be awarded if there is a gold, silver and a bronze medal?

Answer: 10 * 9 * 8.

Explanation: There are 10 choices for gold medal, 9 for silver and 8 for bronze.

Ordered vs. Unordered

In all the examples above, the choices were ordered, meaning that the order of the objects mattered. The rooks were of different colors and so were the medals.

Question: How many ways are there to place 2 white rooks on a 4x4 board so that they do not attack each other?

Answer: 72. Color one of the rooks red. Now there are 144 ways to place the rooks. However, if rooks switch places, in a distinguishable case this leads to a different arrangement of rooks while in this problem it does not. Hence the answer is 144 / 2 = 72.

Combinations - Mathlandia

- There are 10 students in a chess club. How many ways are there to pick two of them and send them to a competition?
- There are 10 cities in Mathlandia. Every two are connected by roads. How many roads are there?

Do you see that this is the same problem?

There are 10 ways to pick the first student and 9 ways to pick the second student.
However, the order in which you picked the students is irrelevant and so you have double counted.
The answer is 10 * 9 / 2 = 45.

In this problem, the order of picked cities does NOT matter.

Question: There are 10 students. How many ways are there to pick 3 of them and send them to a competition?

Solution: There are 10 * 9 * 8 ways to pick the students in an ordered way:
(student_1, student_2, student_3).
How many ways are there to order 3 students?
There are **3!** ways to do so, so you have overcounted by a factor of 6.

Hence there are (10 * 9 * 8) / 3! ways to pick the students.

$\binom{n}{k}$ n-choose-k

Question: There are n students in a classroom. How many ways are there to pick k of them and send them to a competition?

How many ways are there to pick n students **"in order"** ?

$$n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot (n-k+1)$$

How many ways to permute k students? **k!**

Hence there are

$$\frac{n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot (n-k+1)}{k!}$$ ways to arrange the students.

Keep in mind to minimize failure:

There are as many numbers in the numerator as in the denominator!

This expression is frequently written a little differently:
Use the fact that

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \ \ldots \ \cdot (n-k+1) \cdot (n-k)!$$

Hence

$$n \cdot (n-1) \cdot (n-2) \cdot \ \ldots \ \cdot (n-k+1) = \frac{n!}{(n-k)!}$$

Put it all together:
Number of ways to pick k distinguishable objects out of n is:

$$\frac{n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot (n-k+1)}{k!} = \frac{n!}{k!(n-k)!} = \binom{n}{k}$$

This expression is so common in counting that it has a name: "**n choose k**".
In Google Doc insert-equation mode: \choose n k

n-choose-k examples

1. k = 0.
   There is 1 way to pick "no items" out of n. n! / (0! * n!) = 1.
2. k = 1.
   There are n! / [(n-1)! * 1!] = n! / (n-1)! = n ways to pick 1 object out of n.
3. k = 2.
   "n choose 2" = n * (n-1) / 2! = n * (n-1) / 2.
   Is this the correct answer for the Mathlandia problem? Yes.

Question: There are 8 girls and 6 boys on a school math team.

a) How many ways are there to pick a boy and a girl and send them to a competition?
Answer: 8 * 6 = 48.
There are 8 ways to choose one girl out of 8; 6 ways to pick a boy. For each choice of a girl, there are 6 choices of a boy. Total is 6 * 8 = 48.

b) How many ways are there to pick 2 boys and 2 girls and send them to a competition?
Answer: there are "8 choose 2" = 8 * 7 / (1 * 2) = 28 ways to pick girls.
"6 choose 2" = 6 * 5 / 2 = 15 ways to pick boys.
Total of 15 * 28 = 420 ways.

c) How many ways to pick a team of 4 students such that there is at least one boy and at least one girl?
Solution 1:
How many ways to pick 4 students regardless of gender?
"14 choose 4" = (14 * 13 * 12 * 11) / (1 * 2 * 3 * 4) = 7 * 11 * 13 = 1001.
Now, use **negative space**.

How many ways to pick 4 girls and no boys?
"8 choose 4" = (8 * 7 * 6 * 5) / (1 * 2 * 3 * 4) = 2 * 7 * 5 = 70 ways.
How many ways to pick 4 boys and no girls?
"6 choose 4" = 6 * 5 * 4 * 3 / (4 * 3 * 2 * 1) = 15.

Total to pick a team of 4 with at least one boy and one girl:

1001 - 70 - 15 = 916 ways.

Solution 2: Use **casework** on the number of boys picked.

How many ways to pick

3 girls and a boy? “8 choose 3” * 6 = (8 * 7 * 6) / (1 * 2 * 3) * 6 = 8 * 7 * 6 = 336.

3 boys and a girl? 8 * “6 choose 3” = 8 * (6 * 5 * 4 / 6) = 160.

2 boys and 2 girls? “8 choose 2” * “6 choose 2” = 420.

Total is 336 + 160 + 420 = 916.